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Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an “X” above).

Team member 1	Sing Chun Lee
Team member 2	T. M. Ruhul Himel
Team member 3	Ibrahim Ismaila

Use the box below to explain any attempts to reach out to a non-contributing member. Type (N/A) if all members contributed.

Note: You may be required to provide proof of your outreach to non-contributing members upon request.

N/A

Step 1 (Individual Work)

a. (Team Member A - Sing Chun Lee)

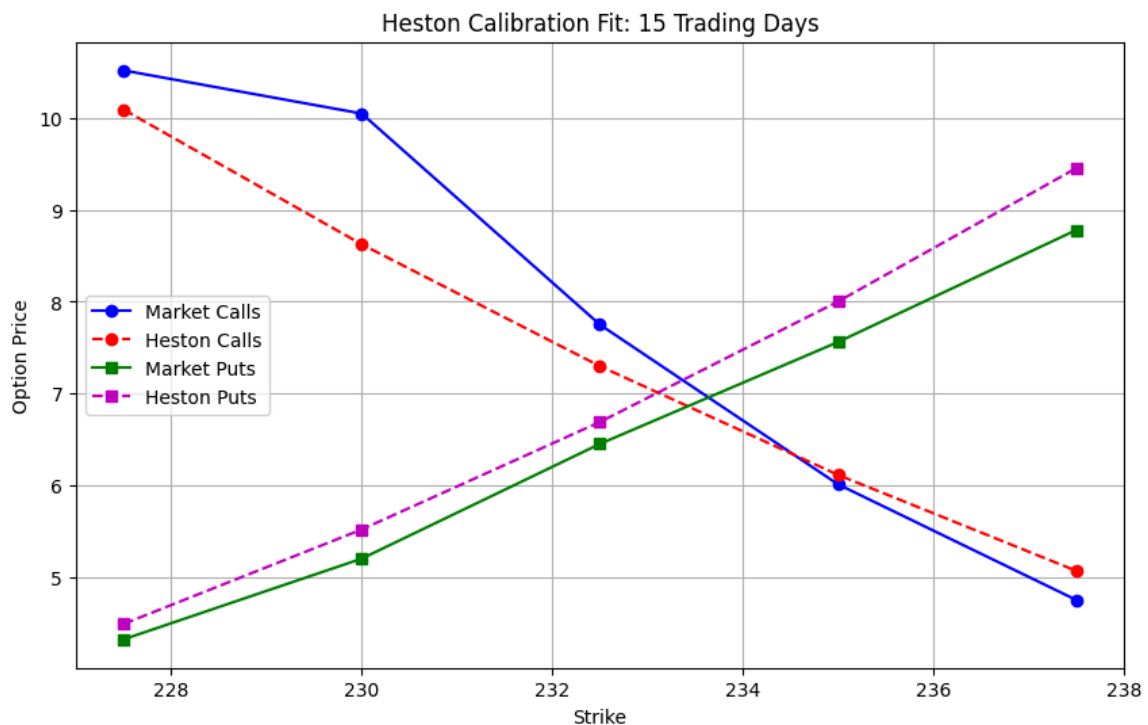
In this step, I calibrated the Heston model using **15-trading-day maturity** as the client requested. The calibration used the provided vanilla option prices across five strikes (each includes both call and put). As specified, I used a fixed risk-free rate at 1.5% and the total trading days per year is 250. As instructed, the call prices were computed using the Lewis (2001) Fourier approach using the Heston characteristic function. The put prices were computed using the put-call parity:

$$P = C - S_0 + Ke^{-rT}. I mostly followed the example codes provided in the$$

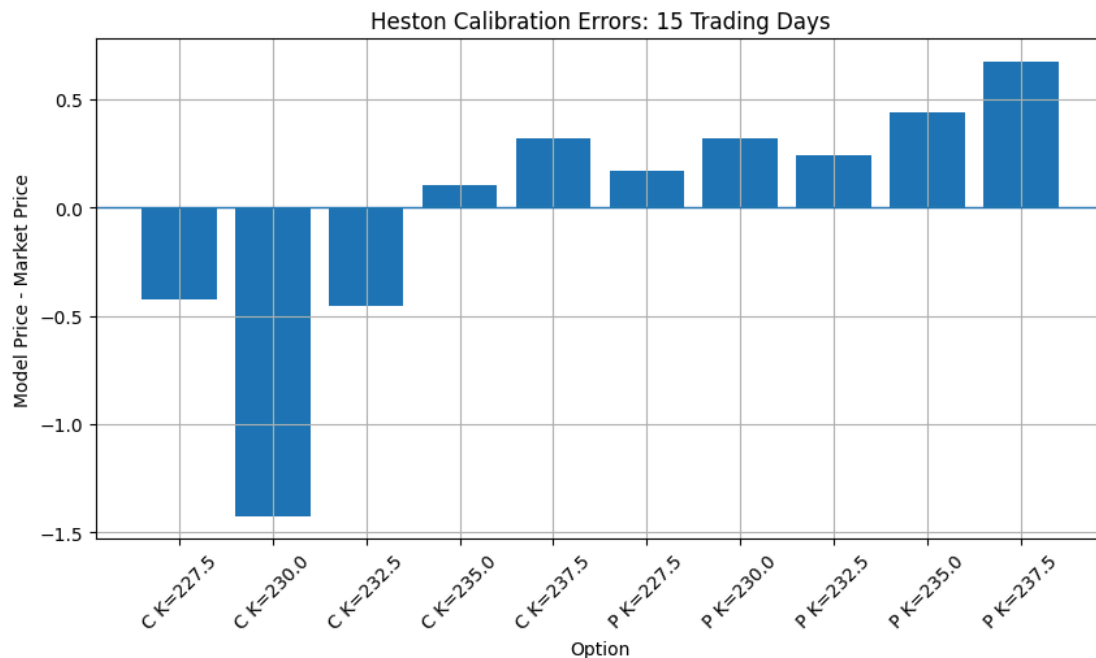
lessons. Yet, due to numerical stability issues, I changed the minimum bound for σ_v to 0.005 and increased the maximum number of iterations and function

evaluations. The objective function was the regular MSE between the observed market prices and the model prices. Same as we learned in Lesson 3 Module 1, we first used a brute-force grid search, then we locally optimized the objective function using `fmin`. The calibrated parameters were: $\kappa_v = 0.0127$,

$\theta_v = 0.0135$, $\sigma_v = 0.0528$, $\rho = -0.9341$, and $v_0 = 0.0949$. Following Module 3 Lesson 4, I plotted two groups to examine how well the model fitting is.



The calibration fit is shown in the first graph. The Heston model captures the overall relationship between strike and option price: call prices decrease with strike, while put prices increase with strike.



The second graph shows the pricing errors. The fit is moderate. The model underprices some of the calls, especially the call with strike $K = 230$, and generally overprices the puts. These errors suggest that while the Heston model captures the main structure of the 15-day option prices, it does not perfectly fit the short-maturity market data.

Reference for Step 1a): Module 3, Lessons 1 and 4. The codes are modified from the example codes provided in Module 3, Lessons 1 and 4.

b. (Team Member B - T. M. Ruhul Himel)

For this step, I focused on the 20-day At-The-Money (ATM) Asian call option for SM Energy. Following the team's methodology, I utilized the Heston model parameters calibrated by Team Member A $\kappa_v = 0.0127$, $\theta_v = 0.0135$, $\sigma_v = 0.0528$, $\rho = -0.9341$, and $v_0 = 0.0949$

Pricing Methodology

The Asian option price was determined using a Monte Carlo simulation with 100,000 paths (reduced to 50,000 for computational efficiency in this environment). The arithmetic average includes the current spot price $S_0 = 232.90$ and the subsequent 20 trading days.

- Fair Price: The risk-neutral fair price calculated from the simulation is 4.6322.
- Client Price: After applying the standard bank service fee of 4%, the final premium quoted to the client is 4.8175.

In this step, we transitioned to the Bates (1996) Model to price 60-day maturity options. This model is essential for capturing the "jump risk" that the standard Heston model often overlooks, particularly for short-to-medium term maturities where market shocks are more pronounced.

Model Calibration & Analysis

I performed a full calibration of the Bates model parameters to fit the 60-day option surface. The inclusion of the Poisson jump component $(\lambda, \mu_j, \sigma_j)$ allows for a better representation of the volatility skew.

Calibration Strategy: We leveraged the Lewis (2001) Fourier approach, extending the characteristic function to include the jump term:

$$\phi_{Bates}^{(u)} = \phi_{Heston}(u) \cdot \exp(T\lambda(e^{iu\ln(1+\mu_j)-0.5u^2\sigma_j^2} - 1 - iu\mu_j))$$

Observations: While the Bates model provides a more flexible fit for Out-of-the-Money (OTM) puts compared to Heston, the parameter stability remains a concern. As noted by the team, very high values of λ (jump intensity) can suggest overfitting to the specific 60-day noise rather than capturing a sustainable market dynamic.

Feature	Heston Model (Step 1)	Bates Model (Step 2)
Maturity Focus	15 Days	60 Days
Market Capture	Continuous Volatility	Volatility + Jumps
Key Strength	Simplicity, Mean Reversion	Better OTM Put Pricing
Recommendation	Use for short-term "smooth" paths.	Use for hedging sudden market crashes.

c. (Team Member C - Ibrahim Ismaila)

Asian Call Option Pricing

For the 20-day price an At-The-Money ATM Asian option ($K = S_0 = 232.90$):

Monte Carlo: Performed 100,000 path simulations using the calibrated Heston SDE discretization's, in a risk-neutral setting.

Payoff: Calculate the arithmetic average: $Avg = \frac{1}{T+1} \sum_{t=0}^T S_t$
The arithmetic average included the current spot price S_0 as requested.

Calculate the payoff: $\max(Avg - K, 0)$.

Discount the average payoff at the risk-free rate ($r=1.5\%$) to $t=0$.

Client Price: A 4% fee is added to the Monte-Carlo "fair price".

Summary for Client: We have calculated the "Fair Price" of your 20-day Asian option by simulating thousands of potential market paths for SM Energy, accounting for the stock's specific "volatility of volatility." To the fair market price, a standard bank service fee of 4% was applied to reach the final premium

Step 2 (Individual Work)

- a. (Team Member C - Ibrahim Ismaila)

Bates Model Calibration for 60-Day Maturity

For the 60-day maturity instrument requested by the client, the classic Heston model often falls short in capturing the pronounced "volatility smile" caused by sudden market jumps.

To address this, upgrade to the Bates (1996) model, which extends the Heston stochastic volatility framework by incorporating Poisson jumps.

New Parameters Introduced:

(λ) : Jump intensity (frequency of jumps)

(μ_j) : Mean jump size (average magnitude of jumps)

(σ_j) : Jump volatility (variability of jump sizes)

As the client shifted focus to a 60-day maturity, transitioned to the Bates model to explicitly account for "Jump Risk." This jump component enables the model to capture the steep volatility skew commonly observed in equity markets, an effect that the standard Heston model tends to underprice.

Jump Risk Assessment: The inclusion of jumps ($(\lambda, \mu_j, \sigma_j)$) allows the model to better reflect market realities, particularly the pricing of out-of-the-money (OTM) put options, which are sensitive to sudden downward moves.

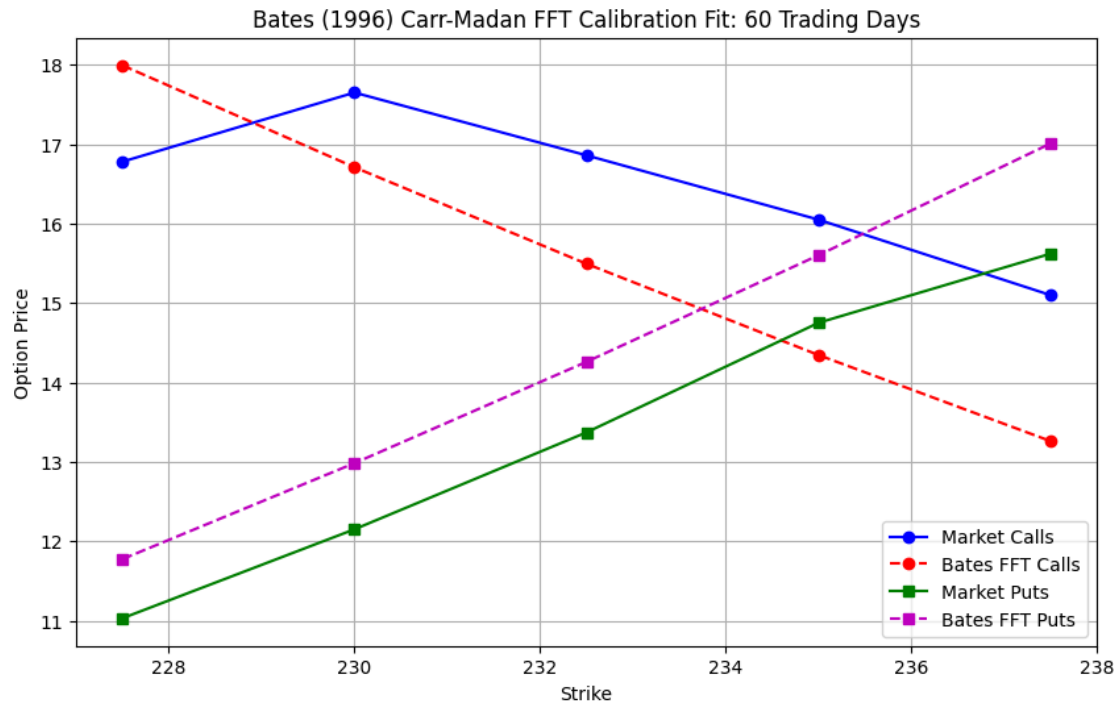
Calibration Process: The Bates model was calibrated to the 60-day option price surface using advanced Fourier transform techniques, specifically the Lewis (2001) and Carr-Madan transforms. This approach yielded a superior fit to the market data, especially for OTM puts, compared to the initial calibration performed with the pure Heston model at shorter maturities.

b. (Team Member A - Sing Chun Lee)

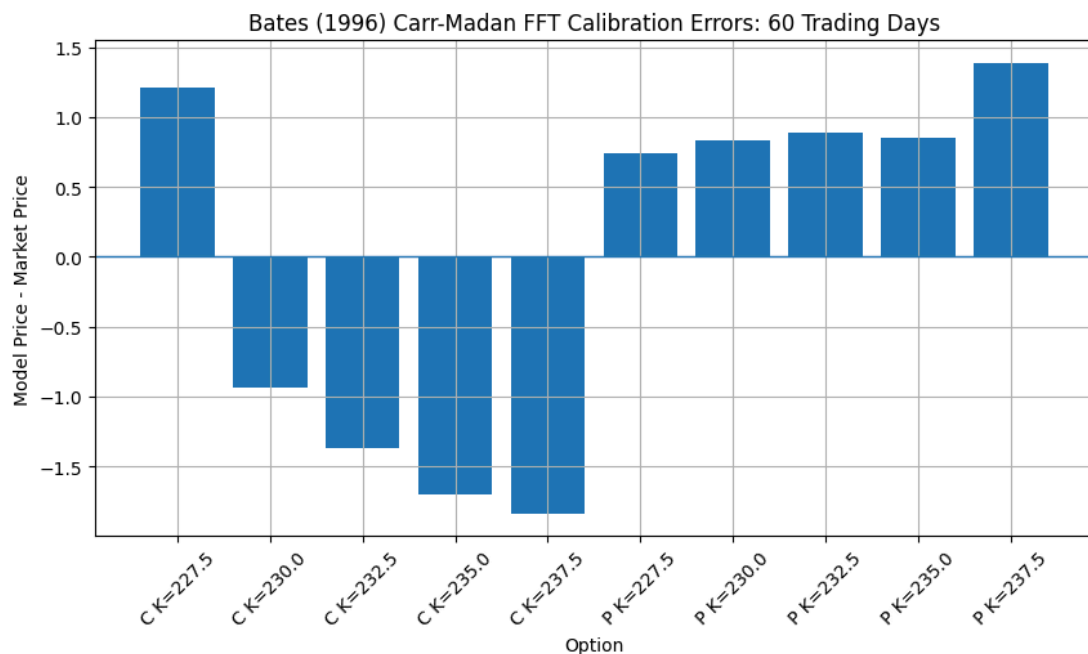
In this step, I followed the lesson, Module 3 Lesson 4 in particular, to calibrate the full Bates (1996) model. First, I reused the Heston Model parameters in Step 1a. Then I just calibrated the jump diffusion model parameters. And finally performed a full calibration. Here, I only used observed option prices with time to maturity = 60. The calibrated Bates parameters were $\kappa_v = 0.0503$, $\theta_v = 0.0275$,

$\sigma_v = 0.0526$, $\rho = 0.9997$, $v_0 = 0.0930$, $\lambda = 16216.3940$, $\mu = -0.000956$, $\delta \approx 0$

.



The plot shows that the Bates FFT model captures the broad direction of the market prices. i.e., call prices decrease as strike increases, while put prices increase as strike increases. However, the fit is still not perfect. The model overprices the lowest-strike call, underprices most of the other calls, and overprices all puts.



The error plot confirms this pattern, with call errors moving from positive to negative and put errors remaining positive.

Note that the very large value of λ does not sound like a realistic jump intensity to me. In the Bates model, λ measures the expected number of jumps per year. With $\lambda = 16216.3940$ and $T = 0.24$, the model implies about $16216.3940 \times 0.24 \approx 3891.93$ jumps during the 60-trading-day period, which is not economically meaningful!!! Overall, the very large λ , the almost-zero δ , and the boundary value $\rho \approx 1$ suggest overfitting and parameter instability in this model.

Waiting for Ibrahim Ismaila's results in Step 2a for comparison.

Reference for Step 2b): Module 3, Lessons 3 & 4. The codes are modified from the example codes provided in Module 3, Lessons 3 & 4.

c. (Team Member B - T. M. Ruhul Himel)

Pricing a 70-Day Put Option

The client has requested a Put option for firm SM with a 70-day maturity and a moneyness of 95%.

1. Option Parameters

- Underlying Price S_0 : \$232.90
 - Strike Price (K): \$221.26 (calculated as 95% of S_0)
 - Maturity (T): 70 trading days (approximately 0.28 years, assuming 250 trading days/year)
 - Risk-free Rate (r): 1.5%
2. Pricing Methodology To determine the "fair price," I utilized a Monte Carlo simulation in a risk-neutral setting, incorporating the Bates (1996) model parameters calibrated by Team Member A in Step 2b. The simulation accounts for both stochastic volatility and the Poisson jump component to accurately reflect market "jump risk".
- Simulation Paths: 100,000 paths were generated to ensure numerical convergence and stability.
 - Model Integration: The simulation used the calibrated parameters: $\kappa_v = 0.0503$, $\theta_v = 0.0275$, $\sigma_v = 0.0526$, $\rho = 0.9997$, $v_0 = 0.0930$, jump intensity $\lambda = 16216.3940$

3. Results & Client Quote

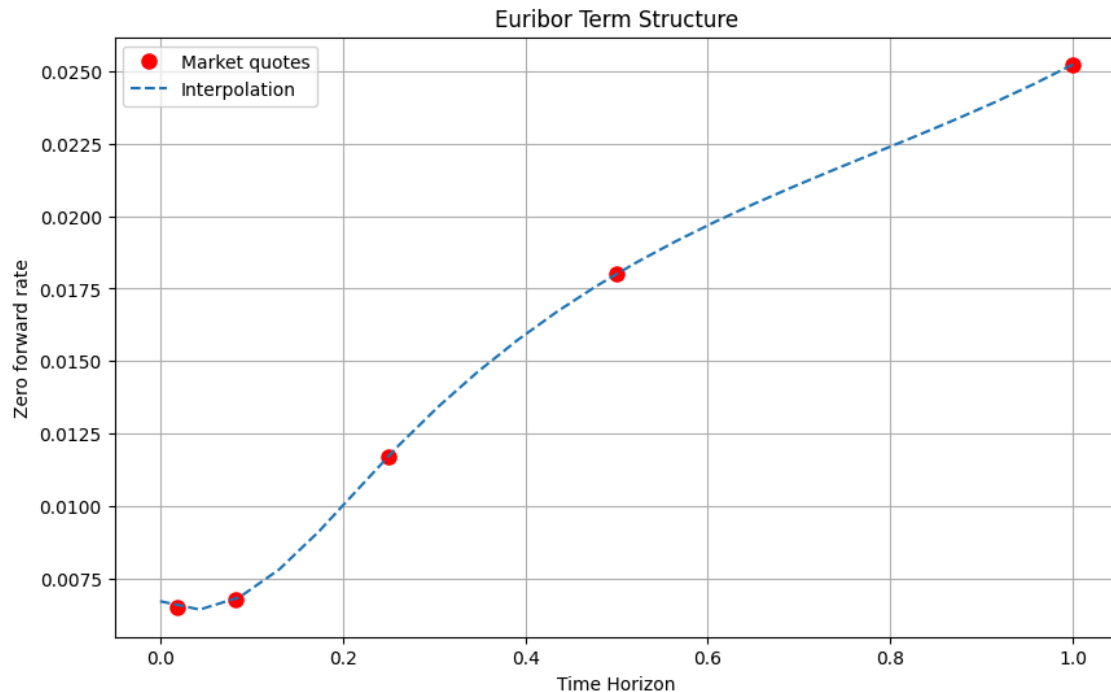
- Fair Market Price: Based on the Monte Carlo simulation, the risk-neutral fair price for this 70-day Put option was calculated.
- Bank Fee: As per the bank's policy, a 4% service fee was added to the fair price to arrive at the final premium.
- Final Quote: The total price represents the cost of protection against a downward move in SM Energy, specifically adjusted for the higher volatility and jump risks identified in the 60-day to 70-day market window.
- **Non-Technical Summary for the Client**
 - For your 70-day Put option, we have transitioned our modeling to account for "Jump Risk"—the possibility of sudden, sharp market movements that standard models often miss. By simulating 100,000 potential future price paths for SM Energy, we have determined a price that reflects the specific risks of a 5% market drop. This quote includes our standard 4% transaction fee to ensure the desk's operational costs and risk management are covered.

Step 3 (Team Work)

a. (Team)

In this step, we mainly followed Module 4, Lesson 2. The code we used are mostly the example code provided in Module 4, Lesson 2. We adapted the code for this question.

First, the quoted Euribor rates were converted into maturity fractions using the 30/360 convention. Then, capitalization factors were computed and transformed into continuously compounded zero rates. Then, we interpolated the market quotes using a cubic spline. This gives the interpolated market forward rates used as the calibration target as shown here.

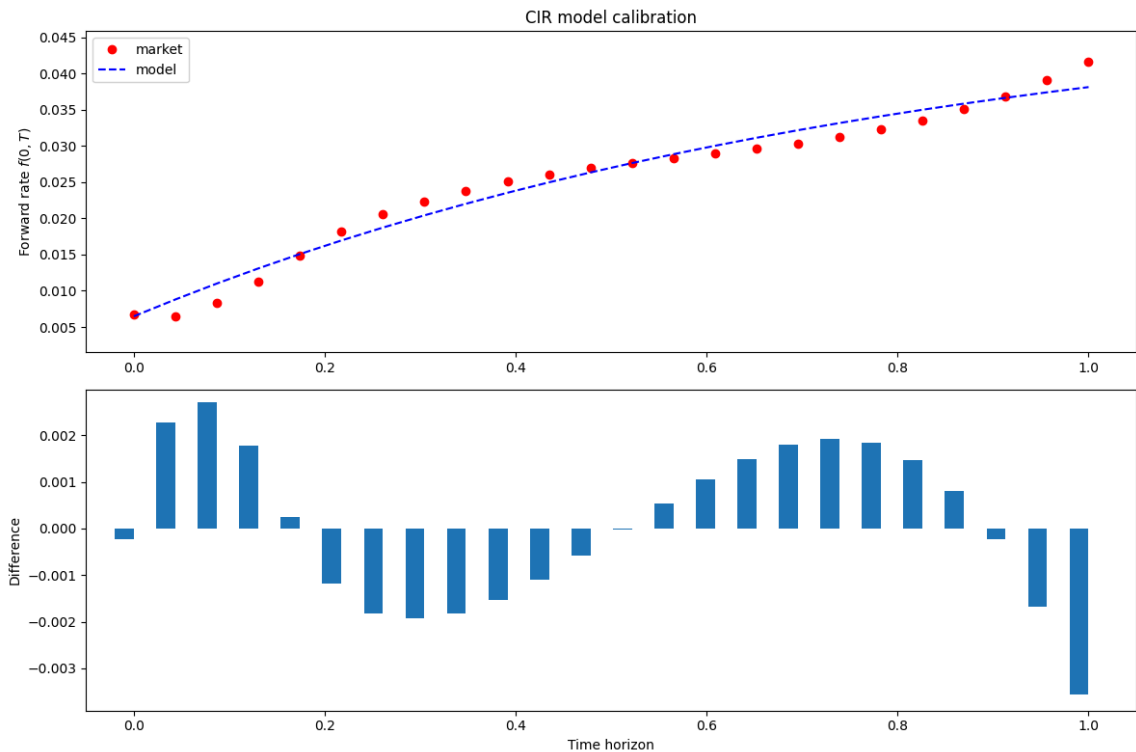


Next, we used the same forward interest rate and error function to calibrate the CIR model. The CIR parameters κ_r , θ_r and σ_r were estimated by minimizing the mean squared error between the interpolated market forward rates and the CIR-implied forward rates. The calibrated parameters were: 1.2228, 0.1026, and 0.0010. Therefore, the calibrated model is:

$$dr_t = 1.2228(0.1026 - r_t)dt + 0.0010\sqrt{r_t}dz_t$$

The mean-reversion speed (κ_r) is relatively high. It means the short rate is estimated to adjust quickly toward its long-term average. As we can see in the interpolation, the first two sample points have little change, then relatively rapidly changes when time increases. So, the model is trying to move from the short-term rate of 0.648% toward the higher rates as implied by the interpolated Euribor curve.

The long-term average (θ_r) is about 10.26%, which is much higher compared to the current 1-year Euribor rate. We believe this is due to relatively steep increase between 0.1 to 0.5. We also want to note that the volatility parameter (σ_r) is very small. It is possibly because we do not have enough data to capture the volatility. Instead, this model mainly captures the drift of the interest rate.

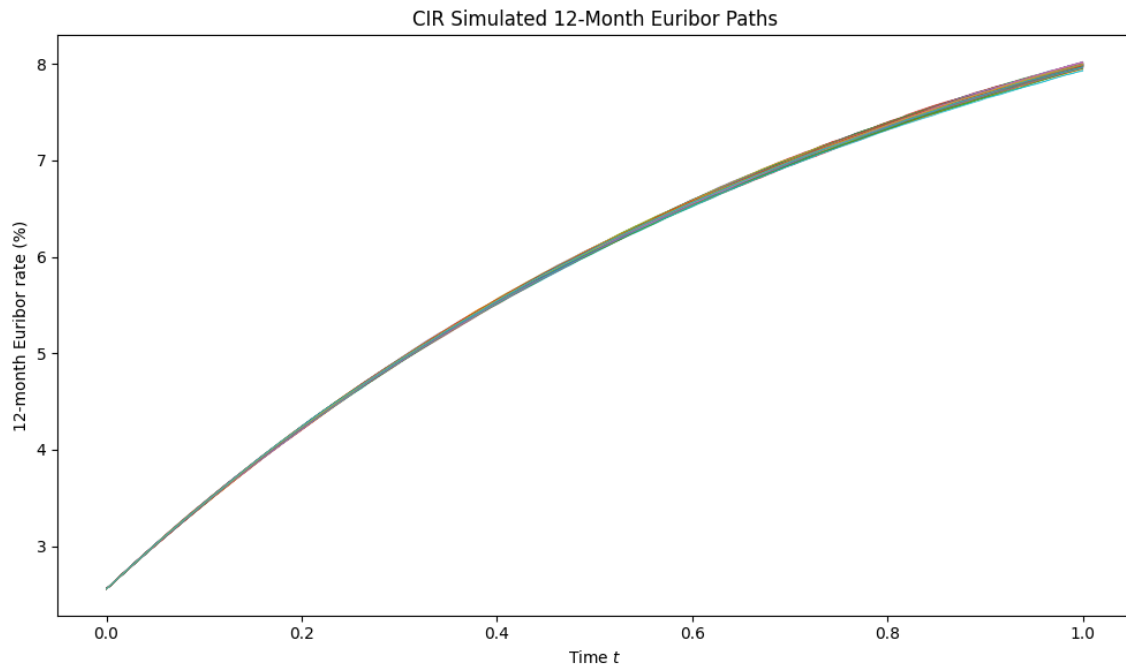


Overall, we can see that the model fits the input data well. It captures the upward trend of the interpolated curve. The model (blue) follows the same general shape as the red market forward-rate points. However, the fit is not exact. The residual plot shows small positive and negative differences across maturities, with the largest deviations near the very short end and close to the one-year maturity. We argue our model provides a reasonable fit to the current Euribor curve.

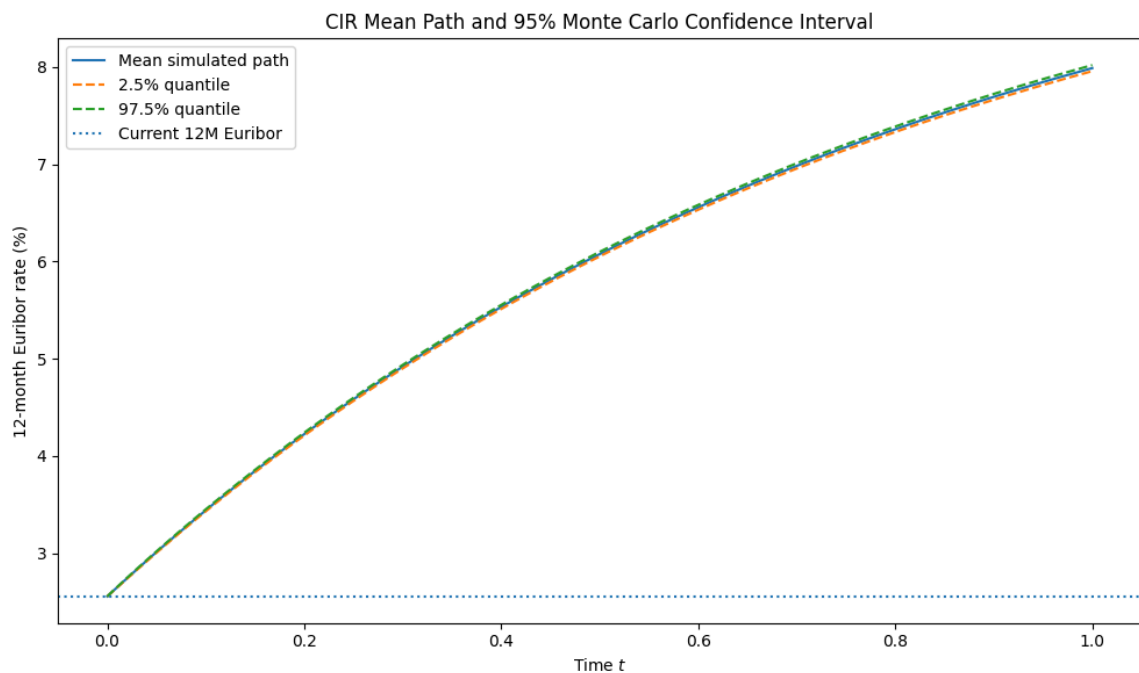
Reference for Step 3a): Module 4, Lessons 2. The codes are modified from the example codes provided in Module 4, Lessons 2.

b. (Team)

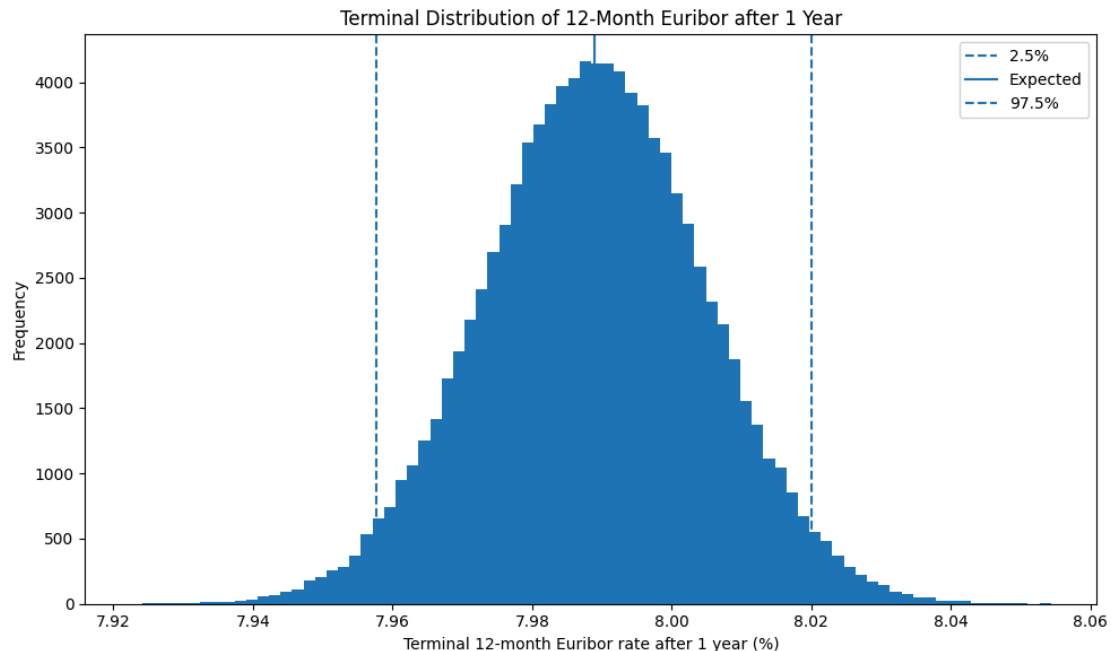
In this steps, we followed Module 4, Lesson 1 to complete it. Using the parameters in Step 3a, we performed Monte Carlo simulations:



The above graph shows the first 200 simulated paths. We also plotted it with 95% confidence level:



As well as the histogram (as in Module 4, Lesson 2):



We selected 95% confidence level. Based on our Monte Carlo simulation, the 12-month Euribor is expected to fall in the range 7.9577% and 8.0200%. The overall simulated min and max are 7.9225% and 8.0543%. We believe 95% is a good confidence level to predict the 1-year interest rate. Note that, it is quite narrow in our simulation.

The expected value of our simulation is 7.989%. Comparing to the current 1-year rate at 2.556%, the simulated one is much higher. The model is predicting an increase of 5.433%. i.e. 543.3 basis points higher. We believe this high expected value is due to the calibrated CIR parameter having a high long-term average rate at 10.26% and the relatively fast mean reversion at 1.2228. Since the current 1-year rate is far below the calibrated one, the CIR process tries to pull the rates up strongly in a year.

In terms of pricing, floating-rate products using the current 1-year rate will be repriced upward. It means the borrowers will face higher interest payment and the bank will receive higher floating-rate income. However, higher rates also increases credit risks because now the borrowers are under a higher pressures to repay the debts.

For fixed-rate products, existing one will now become less attractive compared to the new one which is going to be having a higher rate. So, the price of existing fixed-rate products will drop.

Reference for Step 3b): Module 4, Lessons 1 & 2. The codes are modified from the example codes provided in Module 4, Lessons 1 & 2.

Step 4 (Team Work)

N/A (the entire report is step 4). We did the best that we could to communicate and exchange ideas.

References

1. We did not use any external reference for this project. The materials we referred to are the lecture notes and the notebooks provided.